

CALCULUS I

Chapter 1: FUNCTIONS OF ONE VARIABLE

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SEQUENCES

Definition 1: A **sequence** can be thought of as a list of numbers written in a definite order:

$$a_1, a_2, a_3, a_4, \dots, a_n, \dots \quad (1)$$

The number a_1 is called the **first term**, a_2 is the **second term**, and in general a_n is the n^{th} **term**. We will deal exclusively with **infinite sequences** and so each term a_n will have a successor a_{n+1} . The sequence (1) is also denoted by $(a_n)_{n=1}^{\infty}$, or shortly, (a_n) .

Example 1:

- $\left(\frac{n}{n+1}\right)_{n=1}^{\infty} = \frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{6}, \frac{6}{7}, \dots$
- $((-1)^{n-1})_{n=1}^{\infty} = 1, -1, 1, -1, 1, -1, \dots$
- $\left(\frac{\sin n}{n}\right)_{n=1}^{\infty} = \frac{\sin 1}{1}, \frac{\sin 2}{2}, \frac{\sin 3}{3}, \frac{\sin 4}{4}, \frac{\sin 5}{5}, \dots$
- $(aq^n)_{n=0}^{\infty} = a, aq, aq^2, aq^3, aq^4, aq^5, \dots$
- $\left(\frac{n^2}{2^n}\right)_{n=1}^{\infty} = \frac{1}{2}, \frac{4}{4}, \frac{9}{8}, \frac{16}{16}, \frac{25}{32}, \frac{36}{64}, \dots$

Definition 2: A sequence (a_n) has the **limit** L and we write

$$\lim_{n \rightarrow \infty} a_n = L \quad \text{or} \quad a_n \rightarrow L \text{ as } n \rightarrow \infty$$

if for every $\varepsilon > 0$ there is a corresponding integer N such that if $n > N$ then $|a_n - L| < \varepsilon$. In this case, we say that the sequence is **convergent**; otherwise, it is **divergent**. Mathematically, we can write

$$\lim_{n \rightarrow \infty} a_n = L \Leftrightarrow \forall \varepsilon > 0, \exists N > 0, \forall n > N, |a_n - L| < \varepsilon$$

Example 2: Show that $\lim_{n \rightarrow \infty} \frac{n}{n+1} = 1$.

Solution : We have: $a_n = \frac{n}{n+1}$ and $L = 1$. In Definition 2, we begin with the last inequality:

$$|a_n - L| < \varepsilon \Leftrightarrow \left| \frac{n}{n+1} - 1 \right| < \varepsilon \Leftrightarrow \frac{1}{n+1} < \varepsilon$$

So, if we choose $N = \left\lceil \frac{1}{\varepsilon} \right\rceil$, then for all $n > N$, we have

$$\left| \frac{n}{n+1} - 1 \right| < \varepsilon. \text{ It means } \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1.$$

Note that $[x]$ is an integer part of a number x

LIMIT LAWS FOR SEQUENCES

If (a_n) , (b_n) , and (c_n) are convergent sequences and C is a constant, then

$$(1) \lim_{n \rightarrow \infty} (a_n + b_n) = \lim_{n \rightarrow \infty} a_n + \lim_{n \rightarrow \infty} b_n$$

$$(2) \lim_{n \rightarrow \infty} (C \cdot a_n) = C \cdot \lim_{n \rightarrow \infty} a_n$$

$$(3) \lim_{n \rightarrow \infty} (a_n \cdot b_n) = \lim_{n \rightarrow \infty} a_n \cdot \lim_{n \rightarrow \infty} b_n$$

$$(4) \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \frac{\lim_{n \rightarrow \infty} a_n}{\lim_{n \rightarrow \infty} b_n} \quad b_n \neq 0 \text{ and } \lim_{n \rightarrow \infty} b_n \neq 0$$

$$(5) \text{ If } \lim_{n \rightarrow \infty} |a_n| = 0, \text{ then } \lim_{n \rightarrow \infty} a_n = 0.$$

$$(6) \text{ If } a_n \leq b_n \leq c_n \text{ and } \lim_{n \rightarrow \infty} a_n = \lim_{n \rightarrow \infty} c_n = L, \text{ then } \lim_{n \rightarrow \infty} b_n = L. \text{ (Squeeze Theorem)}$$

SOME USEFUL LIMITS

$$(1) \lim_{n \rightarrow \infty} q^n = 0, \quad (|q| < 1)$$

$$(2) \lim_{n \rightarrow \infty} \sqrt[n]{n} = 1, \quad \lim_{n \rightarrow \infty} \sqrt[n]{a} = 1, \quad (a > 0)$$

$$(3) \lim_{n \rightarrow \infty} \frac{n^\alpha}{a^n} = 0, \quad (a > 1)$$

$$(4) \lim_{n \rightarrow \infty} \frac{a^n}{n!} = 0, \quad (a > 0)$$

$$(5) \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e$$

$$(6) \lim_{n \rightarrow \infty} \frac{a_0 n^k + a_1 n^{k-1} + \dots + a_n}{b_0 n^m + b_1 n^{m-1} + \dots + b_n} = \begin{cases} 0, & k < m \\ \frac{a_0}{b_0} & k = m \\ \infty & k > m \end{cases}$$

Definition 3: Given a sequence (a_n) . We say

- (a_n) is **increasing** if for all n : $a_n \leq a_{n+1}$
- (a_n) is **decreasing** if for all n : $a_n \geq a_{n+1}$
- (a_n) is **bounded above** if there exists a constant M such that for all n : $a_n \leq M$
- (a_n) is **bounded below** if there exists a constant m such that for all n : $a_n \geq m$

Example 3:

- The sequence $\left(\frac{n}{n+1}\right)$ is increasing and bounded above.
- The sequence $((-1)^{n-1})_{n=1}^{\infty}$ is bounded (above and below) but is not increasing and decreasing.

Theorem 1:

- *Every increasing, bounded above sequence is convergent.*
- *Every decreasing, bounded below sequence is convergent.*

Example 4: Consider the sequence (q^n) where $0 \leq q < 1$.

We see that

- for all n , $q^n > q \cdot q^n = q^{n+1} \Rightarrow (q^n)$ is decreasing.
- for all n , $q^n \geq 0 \Rightarrow (q^n)$ is bounded below.

Therefore, $\exists a$ such that $a = \lim_{n \rightarrow \infty} q^n = \lim_{n \rightarrow \infty} q^{n+1} = qa$.

Since $q \neq 1 \Rightarrow a = 0$.

Example 5: Consider the sequence (a_n) where

$$a_n = \underbrace{\sqrt{2 + \sqrt{2 + \cdots + \sqrt{2}}}}_{n \text{ square root}}$$

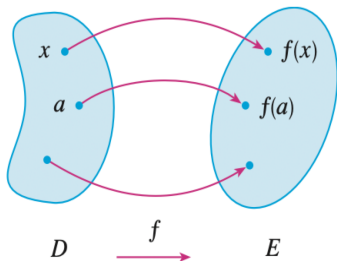
We have $a_{n+1} = \sqrt{2 + a_n}$ and $0 < a_1 = \sqrt{2} < 2$ so by induction, we can prove that $0 < a_n < 2$ for all $n \in \mathbb{N}^* \Rightarrow (a_n)$ is bounded above. On the other hand

$$a_{n+1} - a_n = \sqrt{2 + a_n} - a_n = \frac{2 + a_n - a_n^2}{\sqrt{2 + a_n} + a_n} = \frac{(2 - a_n)(a_n + 1)}{\sqrt{2 + a_n} + a_n}$$

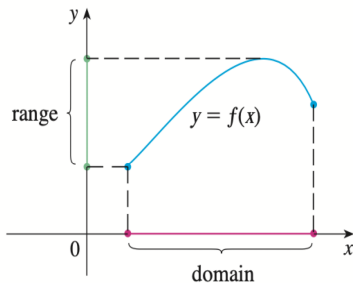
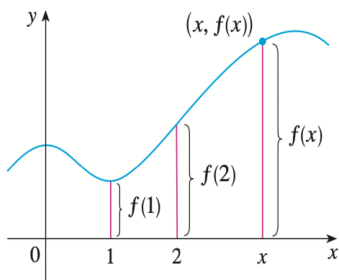
Since $a_n < 2 \Rightarrow (a_n)$ is increasing. Therefore $\exists a : 0 \leq a \leq 2$ and $\lim_{n \rightarrow \infty} a_n = a$. In the equality $a_{n+1} = \sqrt{2 + a_n}$ let $n \rightarrow \infty$, we obtain $a = \sqrt{2 + a} \Rightarrow a = 2$.

DEFINITION OF A FUNCTION

Definition 4: A **function** f is a rule that assigns to each element x in a set D exactly one element, called $f(x)$, in a set E . The set D is called the **domain** of the function. The **range** of f , denoted by R_f , is the set of all possible values of $f(x)$ as x varies throughout the domain. Note that $R_f \subset E$.



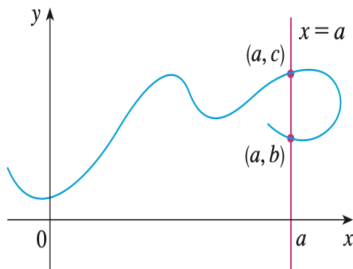
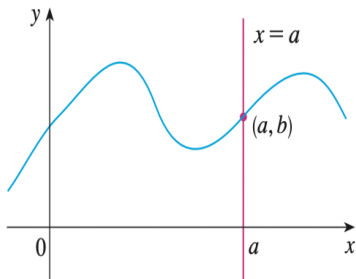
Definition 5: If f is a function with domain D , then its **graph** is the set of ordered pairs $\{(x, f(x)) \mid x \in D\}$. In other words, the graph of f consists of all points (x, y) in the coordinate plane such that $y = f(x)$ and x is in the domain of f . The graph of a function is a curve in the xy -plane.



The following test is used to test any curves, which are graphs of functions.

The Vertical Line Test

A curve in the xy -plane is the graph of a function of x if and only if no vertical line intersects the curve more than once.



Four Ways to Represent a Function

There are four possible ways to represent a function:

- **verbally** (by a description in words). For example, the area of a circle as a function of its radius.

- **algebraically** (by an explicit formula). For example,

$$A = \pi r^2$$

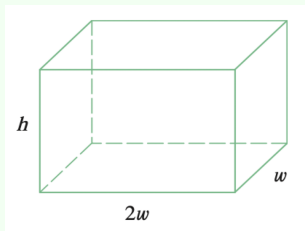
- **numerically** (by a table of values). For example,

r	1	2	3	...
A	π	4π	9π	...

- **visually** (by a graph). We can sketch a graph of A (half a parabola).

Because a circle has to have a positive radius, the domain is $\{r \mid r > 0\} = (0, \infty)$, and the range is also $(0, \infty)$.

Example 6:



A rectangular storage container with an open top has a volume of $10m^3$. The length of its base is twice its width. Material for the base costs \$10 per square meter; material for the sides costs \$6 per square meter. Express the cost of materials as a function of the width of the base.

Solution : Let $w, 2w, h$ be the width, length, and height of the container ($w > 0$). The total cost is

$$\begin{aligned} C(w) &= 10(2w^2) + 6[2(wh) + 2(2wh)] \\ &= 20w^2 + 36wh \end{aligned}$$

Since the volume of the container is $10m^3$, we have

$$(w) \cdot (2w) \cdot (h) = 10 \Rightarrow h = \frac{5}{w^2}$$

Finally, we obtain

$$C(w) = 20w^2 + 36w \cdot \frac{5}{w^2} = 20w^2 + \frac{180}{w} \quad (w > 0)$$

Example 7: Find the domain of each function.

(1) $f(x) = \sqrt{x+2}$

(2) $g(x) = \frac{1}{x^2 - x}$

Solution :

(1) The domain of f consists of all values of x such that $x + 2 \geq 0 \Leftrightarrow x \geq -2$, so the domain is the interval $[-2, +\infty)$.

(2) Since $g(x) = \frac{1}{x(x-1)}$ and division by 0 is not allowed, we see that $g(x)$ is not defined when $x = 0$ or $x = 1$. Thus the domain of g is $\{x \mid x \neq 0, x \neq 1\}$.

Example 8: Find the domain and range of the function

$$f(x) = \frac{2x}{x^2 + 1}.$$

Solution : We see that this function is defined for all $x \in \mathbb{R}$,
so $D_f = \mathbb{R}$.

In order to find the range of this function, we use the following inequality

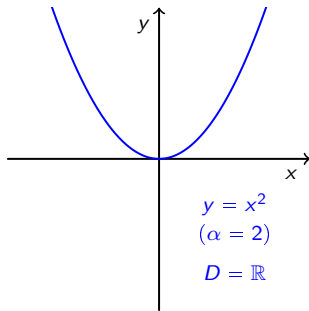
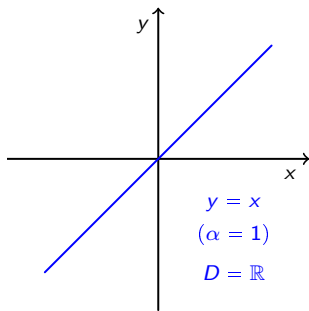
$$\forall x \in \mathbb{R}, \quad -1 \leq \frac{2x}{x^2 + 1} \leq 1$$

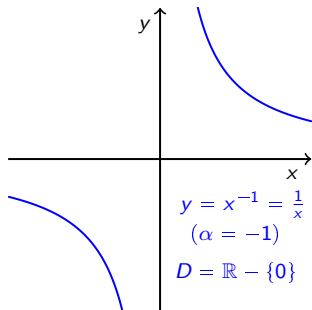
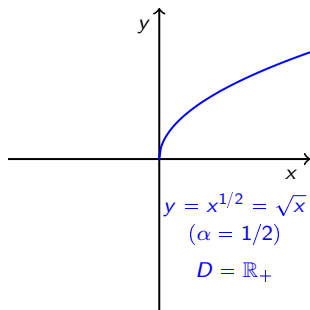
and $f(-1) = -1$, $f(1) = 1$. Finally, $R_f = [-1, 1]$.

THE FUNDAMENTAL FUNCTIONS

Definition 6 (The Power Function): The functions $f(x) = x^\alpha$, where α is a real constant, is called **power functions**.

Their domain and graph are dependent on α . Here are graphs of power functions for some values of α .





Definition 7 (Polynomials): A function P is called a **polynomial** if

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

where n is a non-negative integer and the numbers $a_0, a_1, a_2, \dots, a_n$ are constants called the **coefficients** of the polynomial. The domain of any polynomial is $\mathbb{R} = (-\infty, +\infty)$. If the leading coefficient $a_n \neq 0$, then the **degree** of the polynomial is n .

Example 9:

- A polynomial of degree 1: $P(x) = ax + b$ is called a **linear** function.
- A polynomial of degree 2: $P(x) = ax^2 + bx + c$ is called a **quadratic** function.
- A polynomial of degree 3: $P(x) = ax^3 + bx^2 + cx + d$ is called a **cubic** function.

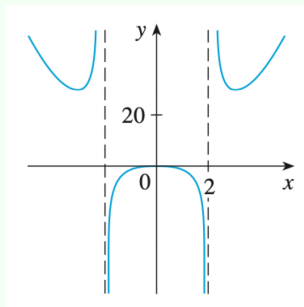
Definition 8 (Rational Functions): A **rational function** f is a ratio of two polynomials: $f(x) = \frac{P(x)}{Q(x)}$, where P and Q are polynomials. The domain consists of all values of x such that $Q(x) \neq 0$.

Example 10:

The function

$$f(x) = \frac{2x^4 - x^2 + 1}{x^2 - 4}$$

is a rational function with domain $\{x \mid x \neq \pm 2\}$.



Definition 9 (Algebraic Functions): A function f is called an **algebraic function** if it can be constructed using algebraic operations (such as **addition**, **subtraction**, **multiplication**, **division**, and **taking roots**) starting with polynomials. Any rational function is automatically an algebraic function.

Example 11: Here are examples of algebraic functions:

- $f(x) = \sqrt{x^2 + 1}$
- $g(x) = \frac{x^4 - 5x^2}{2x + \sqrt{x}} + (x + 3)\sqrt[3]{x - 1}$
- $h(x) = \frac{(x + 2)^3 + 4\sqrt{x}}{(1 + \sqrt{x})^5 + 3x^2}$

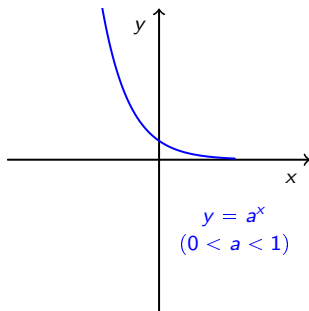
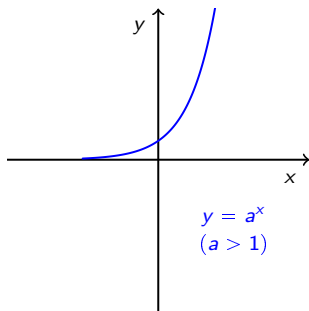
Definition 10 (The Exponential Functions): The function $f(x) = a^x$, where a is a real constant ($a > 0$, $a \neq 1$), is called an **exponential function**.

The exponential function is defined for all $x \in \mathbb{R}$.

Properties of Exponential Functions

- $a^x > 0$ for all $x \in \mathbb{R}$
- $a^0 = 1$
- $a^{x+y} = a^x \cdot a^y$
- $a^{x-y} = \frac{a^x}{a^y} \implies a^{-x} = \frac{1}{a^x}$
- $(a^x)^y = a^{xy}$

When $a = e$ ($e \approx 2.718281828\dots$) we have the most popular function $y = e^x$, that has many applications in all sciences. The graphs of the exponential functions are dependent on a .



Definition 11 (The Logarithmic Functions): The function $f(x) = \log_a x$, where a is a real constant ($a > 0$, $a \neq 1$), is called a **logarithmic function**.

The logarithmic function is defined for all $x > 0$.

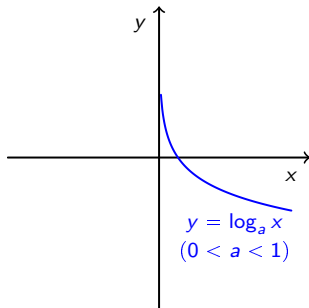
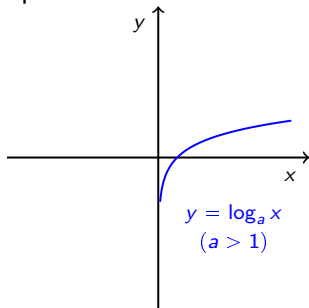
Properties of Logarithmic Functions

- $\log_a 1 = 0$
- $\log_a (xy) = \log_a x + \log_a y$
- $\log_a \frac{x}{y} = \log_a x - \log_a y \implies \log_a \frac{1}{x} = -\log_a x$
- $\log_a x^\alpha = \alpha \log_a x$

When $a = e$ ($e \approx 2.718281828\dots$) we denote $y = \ln x$, called a **natural logarithm** of x , and we have

$$\log_a x = \frac{\ln x}{\ln a}$$

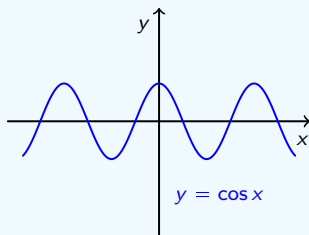
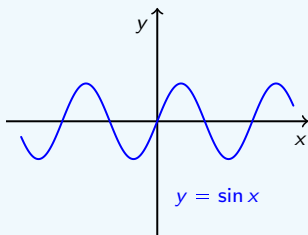
If $a = 10$ we denote $y = \log x$ or $y = \lg x$ and call a **decimal logarithm** of x . The graphs of the logarithmic functions are dependent on a .



Definition 12 (The Trigonometric Functions): We have the following trigonometric functions and their graphs.

$y = \sin x$: its domain is $\mathbb{R}$, its range is $[-1, 1]$.

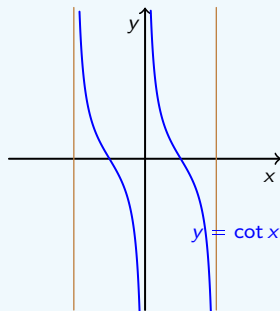
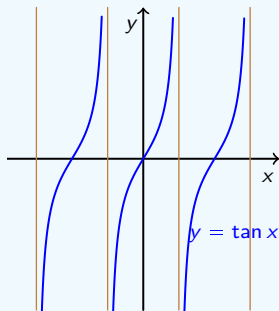
$y = \cos x$: its domain is $\mathbb{R}$, its range is $[-1, 1]$.



Definition 13 (The Trigonometric Functions):

$$y = \tan x = \frac{\sin x}{\cos x} : \text{its domain is } \mathbb{R} - \left\{ \frac{\pi}{2} + k\pi, k \in \mathbb{Z} \right\}.$$

$$y = \cot x = \frac{1}{\tan x} = \frac{\cos x}{\sin x} : \text{its domain is } \mathbb{R} - \{k\pi, k \in \mathbb{Z}\}.$$



Properties of Trigonometric Functions

- $\cos^2 x + \sin^2 x = 1$
- $\sin 2x = 2 \sin x \cos x$
- $\cos 2x = \cos^2 x - \sin^2 x = 1 - 2 \sin^2 x = 2 \cos^2 x - 1.$

$$\Rightarrow \cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}, \quad \sin 2x = \frac{2 \tan x}{1 + \tan^2 x}$$

From this property we also obtain:

$$\cos^2 x = \frac{1 + \cos 2x}{2}; \quad \sin^2 x = \frac{1 - \cos 2x}{2}$$

- $1 + \tan^2 x = \frac{1}{\cos^2 x}$, and from this we have

$$\cos^2 x = \frac{1}{1 + \tan^2 x}; \quad \sin^2 x = \frac{\tan^2 x}{1 + \tan^2 x}$$

- And more

Example 12:

$$\begin{aligned}\cos^4 x &= \left(\frac{1 + \cos 2x}{2} \right)^2 = \frac{1 + 2 \cos 2x + \cos^2 2x}{4} \\&= \frac{1 + 2 \cos 2x + \frac{1 + \cos 4x}{2}}{4} \\&= \frac{3}{8} + \frac{1}{2} \cos 2x + \frac{1}{8} \cos 4x \\ \sin^4 x &= \left(\frac{1 - \cos 2x}{2} \right)^2 = \frac{1 - 2 \cos 2x + \cos^2 2x}{4} \\&= \frac{1 - 2 \cos 2x + \frac{1 + \cos 4x}{2}}{4} \\&= \frac{3}{8} - \frac{1}{2} \cos 2x + \frac{1}{8} \cos 4x\end{aligned}$$

Definition 14 (The Hyperbolic Functions): The following functions are called the **hyperbolic functions**:

(1) The Sine Hyperbolic Function:

$$y = \sinh x = \frac{e^x - e^{-x}}{2}, \quad D = \mathbb{R}, \quad R = \mathbb{R}$$

(2) The Cosine Hyperbolic Function:

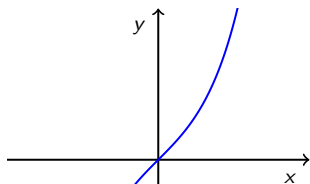
$$y = \cosh x = \frac{e^x + e^{-x}}{2}, \quad D = \mathbb{R}, \quad R = [1, +\infty)$$

(3) The Tangent Hyperbolic Function:

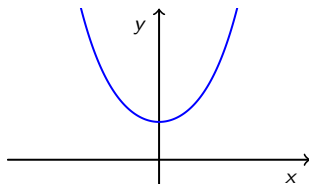
$$y = \tanh x = \frac{\sinh x}{\cosh x} = \frac{e^x - e^{-x}}{e^x + e^{-x}}, \quad D = \mathbb{R}, \quad R = (-1, 1)$$

(4) The Cotangent Hyperbolic Function:

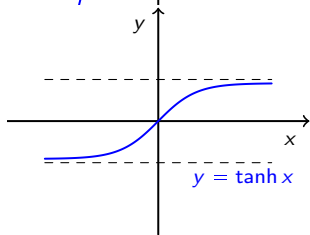
$$y = \coth x = \frac{1}{\tanh x} = \frac{e^x + e^{-x}}{e^x - e^{-x}}, \quad D = \mathbb{R} - \{0\}, \quad R = \mathbb{R} - [-1, 1]$$



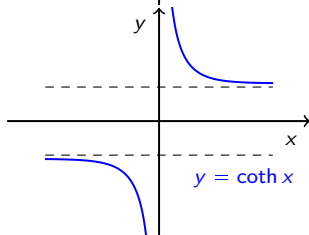
$$y = \sinh x$$



$$y = \cosh x$$



$$y = \tanh x$$



$$y = \coth x$$

Properties of Hyperbolic Functions

- $\cosh^2 x - \sinh^2 x = 1$

$$\begin{aligned}\cosh^2 x - \sinh^2 x &= \left(\frac{e^x + e^{-x}}{2} \right)^2 - \left(\frac{e^x - e^{-x}}{2} \right)^2 \\ &= \frac{e^{2x} + 2 + e^{-2x}}{4} - \frac{e^{2x} - 2 + e^{-2x}}{4} = 1\end{aligned}$$

- $\sinh 2x = 2 \sinh x \cosh x$

$$\begin{aligned}\sinh 2x &= \frac{e^{2x} - e^{-2x}}{2} = \frac{(e^x)^2 - (e^{-x})^2}{2} \\ &= \frac{(e^x - e^{-x})(e^x + e^{-x})}{2} \\ &= 2 \cdot \frac{e^x - e^{-x}}{2} \cdot \frac{e^x + e^{-x}}{2} \\ &= 2 \sinh x \cosh x\end{aligned}$$

Properties of Hyperbolic Functions

- $\cosh 2x = \cosh^2 x + \sinh^2 x$

$$\begin{aligned}\cosh^2 x + \sinh^2 x &= \left(\frac{e^x + e^{-x}}{2} \right)^2 + \left(\frac{e^x - e^{-x}}{2} \right)^2 \\&= \frac{e^{2x} + 2 + e^{-2x}}{4} + \frac{e^{2x} - 2 + e^{-2x}}{4} \\&= \frac{e^{2x} + e^{-2x}}{2} = \cosh 2x\end{aligned}$$

From this we have

- $\cosh 2x = 1 + 2 \sinh^2 x = 2 \cosh^2 x - 1$
- $\cosh^2 x = \frac{1 + \cosh 2x}{2}, \quad \sinh^2 x = \frac{\cosh 2x - 1}{2}$
- And more

Definition 15 (Piecewise Defined Functions): The functions in the following examples are defined by different formulas in different parts of their domains. Such functions are called **piecewise defined functions**.

Example 13: A function f is defined by

$$f(x) = \begin{cases} 1 - x, & \text{if } x \leq -1 \\ x^2, & \text{if } x > -1 \end{cases}$$

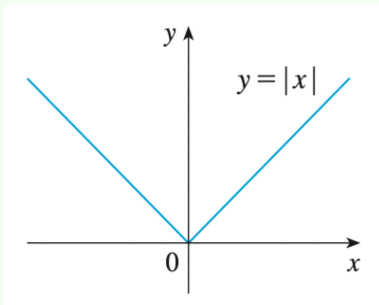
Then

- Since $-2 \leq -1$, we have $f(-2) = 1 - (-2) = 3$
- Since $-1 \leq -1$, we have $f(-1) = 1 - (-1) = 2$
- Since $0 > -1$, we have $f(0) = 0^2 = 0$

Example 14:

The next example of a piecewise defined function is the **absolute value function**

$$f(x) = |x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

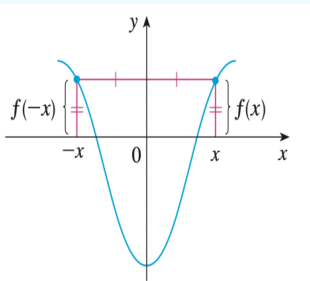


We see that the graph of f coincides with the line $y = x$ to the right of the y -axis and coincides with the line $y = -x$ to the left of the y -axis.

PROPERTIES OF FUNCTIONS

Definition 16 (Even Functions):

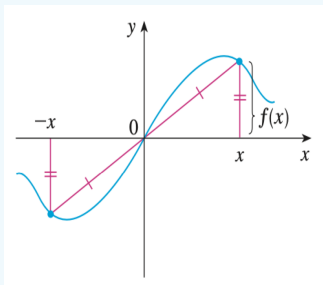
If a function $f(x)$ satisfies $f(-x) = f(x)$ for every number x in its domain, then f is called an **even function**. For instance, the function $f(x) = x^2$ is even because $f(-x) = (-x)^2 = x^2 = f(x)$.



The geometric significance of an even function is that its graph is symmetric with respect to the y -axis (see Figure). This means that if we have plotted the graph of f for $x \geq 0$, we obtain the entire graph simply by reflecting this portion about the y -axis.

Definition 17 (Odd Functions):

If a function $f(x)$ satisfies $f(-x) = -f(x)$ for every number x in its domain, then f is called an **odd function**. For instance, the function $f(x) = x^3$ is odd because $f(-x) = (-x)^3 = -x^3 = -f(x)$.



The geometric significance of an even function is that its graph is symmetric about the y-axis (see Figure). This means that if we have plotted the graph of f for $x \geq 0$, we obtain the entire graph simply by rotating this portion through 180° about the origin.

Example 15: Determine whether each of the following functions is even, odd, or neither even nor odd.

(1) $f(x) = x^5 + x$; (2) $g(x) = 1 - x^4$; (3) $h(x) = 2x - x^2$

Solution :

(1) $f(-x) = (-x)^5 + (-x) = -x^5 - x = -(x^5 + x) = -f(x)$. Therefore f is an odd function.

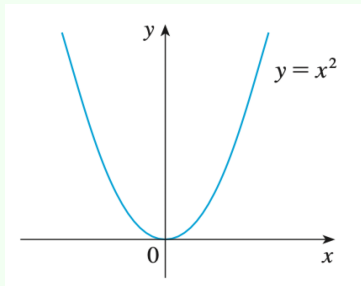
(2) $g(-x) = 1 - (-x)^4 = 1 - x^4 = g(x)$. So g is even.

(3) Since $h(-x) \neq h(x)$ and $h(-x) \neq -h(x)$, we conclude that h is neither even nor odd.

Definition 18 (Increasing and Decreasing Function): A function f is called **increasing** on an interval I if $f(x_1) < f(x_2)$ whenever $x_1 < x_2$ in I . It is called **decreasing** on I if $f(x_1) > f(x_2)$ whenever $x_1 < x_2$ in I .

Example 16:

The function $y = x^2$ is decreasing on $(-\infty, 0)$ and increasing on $(0, +\infty)$.



Definition 19 (Periodic Functions): A function f is called **periodic** if there exists a positive constant τ such that for all x from its domain $f(x + \tau) = f(x)$. The smallest positive value T of all values of τ is call the **period** of this function.

Example 17:

- The functions $y = \sin x$ and $y = \cos x$ are periodic with $T = 2\pi$.
- The functions $y = \tan x$ and $y = \cot x$ are periodic with $T = \pi$.

Example 18:

- The function $y = \sin kx$ is periodic with $T = \frac{2\pi}{k}$.
- The functions $y = \tan \frac{x}{2}$ and $y = \tan \frac{x}{3}$ are periodic with the periods $T = 2\pi$ and $T = 3\pi$, respectively. Then the function $f(x) = \tan \frac{x}{2} + \tan \frac{x}{3}$ is periodic with the period $T = \text{SCM}(2\pi, 3\pi) = 6\pi$.

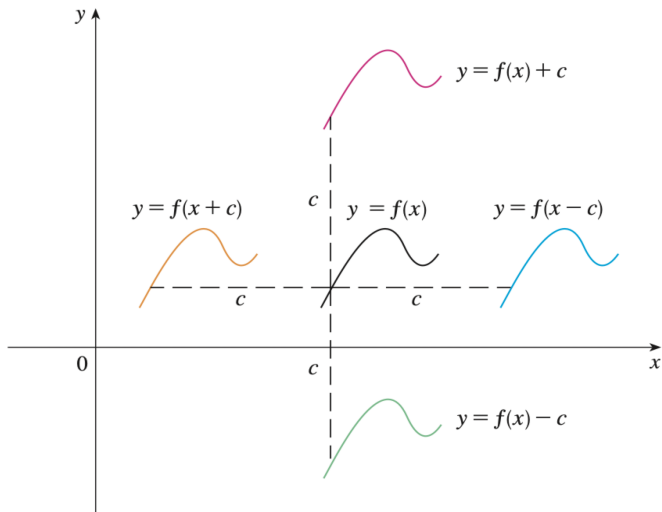
(SCM = Smallest Common Multiple)

TRANSFORMATIONS OF FUNCTIONS

Vertical and Horizontal Shifts

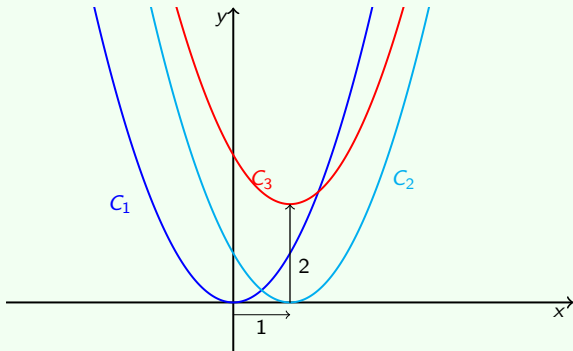
Suppose $c > 0$. To obtain the graph of

- $y = f(x) + c$, shift the graph of $y = f(x)$ a distance c units upward
- $y = f(x) - c$, shift the graph of $y = f(x)$ a distance c units downward
- $y = f(x - c)$, shift the graph of $y = f(x)$ a distance c units to the right
- $y = f(x + c)$, shift the graph of $y = f(x)$ a distance c units to the left



Example 19: Sketch the graph of $y = x^2 - 2x + 3$.

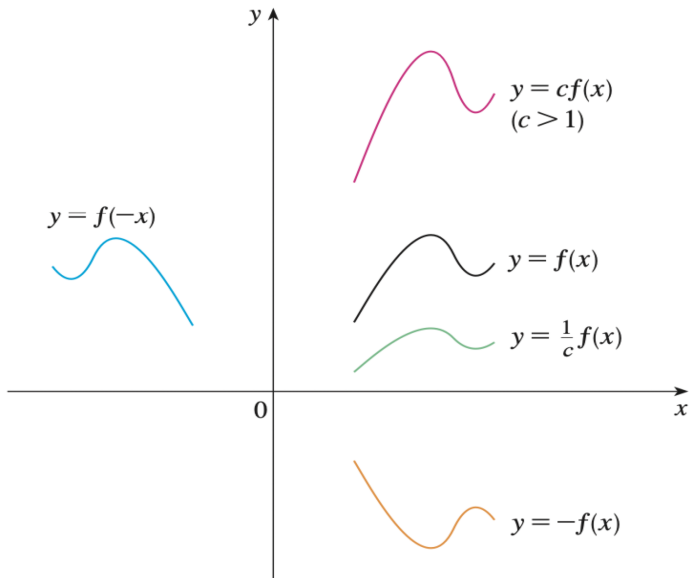
We have $y = (x - 1)^2 + 2$. We begin with the graph of $y = x^2$ (C_1). We shift C_1 a distance 1 units to the right to obtain the graph of $y = (x - 1)^2$ (C_2). Finally, we shift C_2 a distance 2 units upward to obtain the graph of $y = (x - 1)^2 + 2$ (C_3).



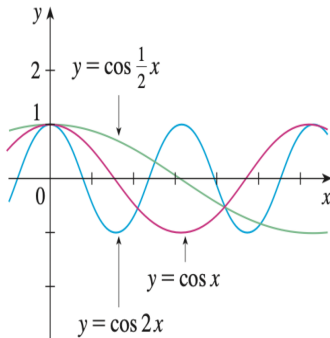
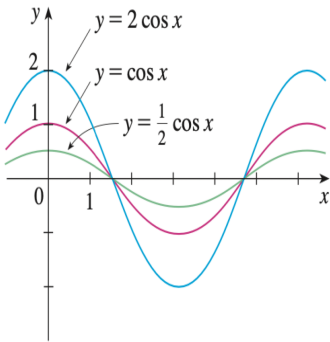
Vertical and Horizontal Stretching and Reflecting

Suppose $c > 1$. To obtain the graph of

- $y = cf(x)$, stretch the graph of $y = f(x)$ vertically by a factor of c
- $y = \frac{1}{c}f(x)$, shrink the graph of $y = f(x)$ vertically by a factor of c
- $y = -f(x)$, reflect the graph of $y = f(x)$ about the x -axis
- $y = f(-x)$, reflect the graph of $y = f(x)$ about the y -axis



Example 20: The following figure illustrates these stretching transformations when applied to the function $y = \cos x$ with $c = 2$ to obtain the graphs of $y = 2 \cos x$, $y = \frac{1}{2} \cos x$, $y = \cos 2x$, and $y = \cos \frac{1}{2}x$.



COMBINATIONS OF FUNCTIONS

Addition, Subtraction, Multiplication, and Division

Two functions f and g can be combined to form new functions $f + g$, $f - g$, fg , and f/g in a manner similar to the way we add, subtract, multiply, and divide real numbers. The sum, difference, product, and quotient functions are defined by

$$(f + g)(x) = f(x) + g(x), \quad (f - g)(x) = f(x) - g(x)$$
$$(fg)(x) = f(x)g(x), \quad (f/g)(x) = \frac{f(x)}{g(x)}$$

Note that if the domain of f is A and of g is B , then the domain of $f \pm g$ is the intersection $A \cap B$ because both $f(x)$ and $g(x)$ have to be defined. The domain of fg is $A \cap B$ too, however, the domain of f/g is $\{x \in A \cap B \mid g(x) \neq 0\}$.

Example 21: The domain of $f(x) = \sqrt{x}$ is $[0, +\infty)$ and the domain of $g(x) = \sqrt{2-x}$ is $(-\infty, 2]$. Therefore

- The domain of $f(x) + g(x) = \sqrt{x} + \sqrt{2-x}$ is $[0, +\infty) \cap (-\infty, 2] = [0, 2]$.
- The domain of $f(x) - g(x) = \sqrt{x} - \sqrt{2-x}$ is also $[0, 2]$.
- The domain of $f(x) \cdot g(x) = \sqrt{x} \cdot \sqrt{2-x} = \sqrt{x(2-x)}$ is also $[0, 2]$.
- The domain of $\frac{f(x)}{g(x)} = \frac{\sqrt{x}}{\sqrt{2-x}} = \sqrt{\frac{x}{2-x}}$ is $[0, 2] - \{2\} = [0, 2)$.

Definition 20 (The Composite Function): Given two functions f and g , the **composite function** $f \circ g$ (also called the **composition**) of f and g is defined by

$$(f \circ g)(x) = f(g(x))$$

The domain of $f \circ g$ is the set of all x in the domain of g such that $g(x)$ is in the domain of f . In other words, $(f \circ g)(x)$ is defined whenever both $g(x)$ and $f(g(x))$ are defined.

Example 22: If $f(x) = x^2$ and $g(x) = x + 3$, find the composite functions $f \circ g$ and $g \circ f$.

Solution : We have

- $(f \circ g)(x) = f(g(x)) = [g(x)]^2 = (x + 3)^2$
- $(g \circ f)(x) = g(f(x)) = f(x) + 3 = x^2 + 3$

We see that, in general, $f \circ g \neq g \circ f$.

Example 23: If $f(x) = \sqrt{x}$ and $g(x) = \sqrt{2-x}$, find each function and its domain: $f \circ g$, $g \circ f$, $f \circ f$, and $g \circ g$.

Solution : We have

- $(f \circ g)(x) = f(g(x)) = \sqrt{g(x)} = \sqrt{\sqrt{2-x}} = \sqrt[4]{2-x}$.
Its domain is $(-\infty, 2]$.
- $(g \circ f)(x) = g(f(x)) = \sqrt{2-f(x)} = \sqrt{2-\sqrt{x}}$. Its domain is $[0, 4]$.
- $(f \circ f)(x) = f(f(x)) = \sqrt{f(x)} = \sqrt{\sqrt{x}} = \sqrt[4]{x}$. Its domain is $[0, +\infty)$.
- $(g \circ g)(x) = g(g(x)) = \sqrt{2-g(x)} = \sqrt{2-\sqrt{2-x}}$.
Its domain is $[-2, 2]$.

It is possible to take the composition of three or more functions. For instance, the composite function $f \circ g \circ h$ is found by first applying h , then g , and then f as follows:

$$(f \circ g \circ h)(x) = f(g(h(x)))$$

Example 24: If $f(x) = \frac{x}{x+1}$, $g(x) = x^5$, and $h(x) = x+2$, then

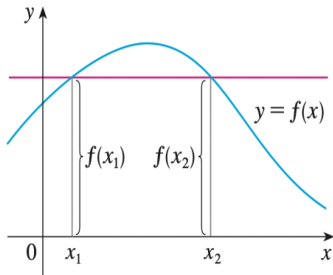
$$\begin{aligned}(f \circ g \circ h)(x) &= f(g(h(x))) = \frac{g(h(x))}{g(h(x)) + 1} = \frac{[h(x)]^5}{[h(x)]^5 + 1} \\ &= \frac{(x+2)^5}{(x+2)^5 + 1}\end{aligned}$$

INVERSE FUNCTIONS

Definition 21: A function f is called a **one-to-one function** if it never takes on the same value twice; that is $f(x_1) \neq f(x_2)$ whenever $x_1 \neq x_2$.

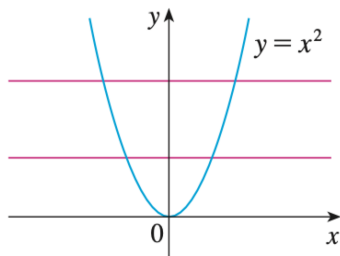
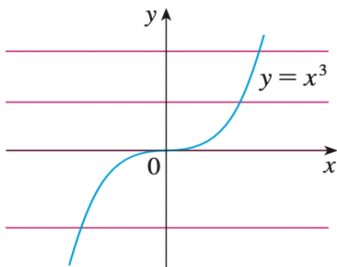
Horizontal Line Test

A function is one-to-one if and only if no horizontal line intersects its graph more than once.



This function is not one-to-one because $x_1 \neq x_2$ and $f(x_1) = f(x_2)$.

Example 25: The function $f(x) = x^3$ is one-to-one on $\mathbb{R}$, because if $x_1 \neq x_2$, then $x_1^3 \neq x_2^3$. The function $g(x) = x^2$ is not one-to-one on $\mathbb{R}$, because $1 \neq -1$ but $g(-1) = g(1) = 1$. However, $g(x)$ is one-to-one on $\mathbb{R}_+ = [0, +\infty)$.



Definition 22: Let f be a one-to-one function with domain D and range E . Then its **inverse function** f^{-1} has domain E and range D and is defined by $f^{-1}(y) = x \Leftrightarrow f(x) = y$ for any y in E .

The following procedure shows you how to find the inverse function of a one-to-one function f :

Step 1 Write $y = f(x)$

Step 2 Solve this equation for x in terms of y (if possible)

Step 3 To express f^{-1} as a function of x , interchange x and y . The resulting equation is $y = f^{-1}(x)$.

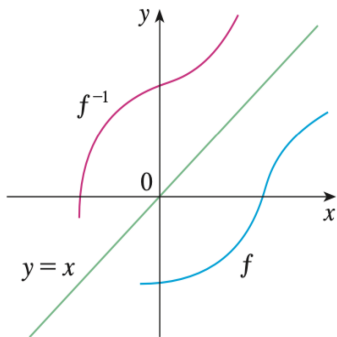
Example 26: Find the inverse function of $f(x) = 2x + 3$.

Solution : According to the above procedure

- We write $y = 2x + 3$.
- Then we solve this equation for x : $x = \frac{y - 3}{2}$.
- Finally, we interchange x and y : $y = \frac{x - 3}{2}$.

Therefore the inverse function is $y = f^{-1}(x) = \frac{x - 3}{2}$.

NOTE



Note that

$$f^{-1}(f(x)) = x$$

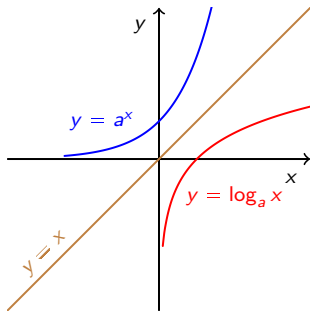
$$f(f^{-1}(x)) = x$$

and the graph of $f^{-1}(x)$ is obtained by reflecting the graph of $f(x)$ about the line $y = x$.

From the elementary functions we have

$$a^{\log_a x} = x \quad \text{and} \quad \log_a a^x = x$$

Then the exponential function $y = a^x$ is the inverse function of the logarithmic function $y = \log_a x$ and vice versa.



Example 27: Find the inverse function of $f(x) = \sinh x$.

Solution : Similarly

- We have $y = \sinh x : \mathbb{R} \longrightarrow \mathbb{R}$ is an 1-1 function.
- To solve this equation for x , we remember that:

$$\sinh x = \frac{e^x - e^{-x}}{2}, \quad \cosh x = \frac{e^x + e^{-x}}{2} = \sqrt{1 + \sinh^2 x}.$$

Therefore,

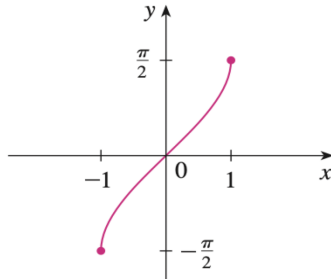
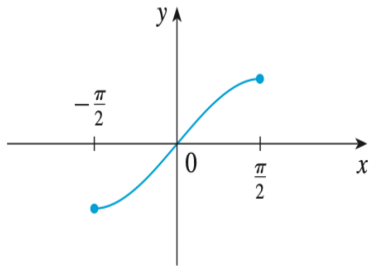
$$\sinh x + \cosh x = y + \sqrt{1 + y^2} = e^x \Rightarrow x = \ln(y + \sqrt{1 + y^2})$$

- Finally, we interchange x and y : $y = \ln(x + \sqrt{1 + x^2})$.

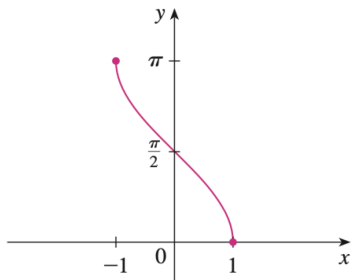
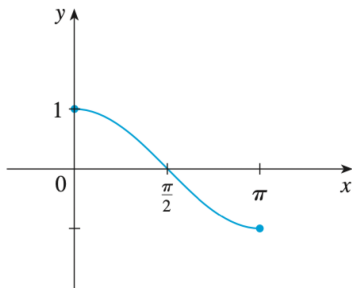
The inverse function of $y = \sinh x$ is

$$y = \sinh^{-1} x = \ln(x + \sqrt{1 + x^2})$$

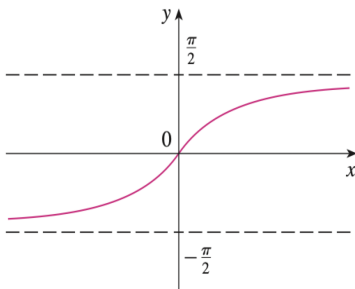
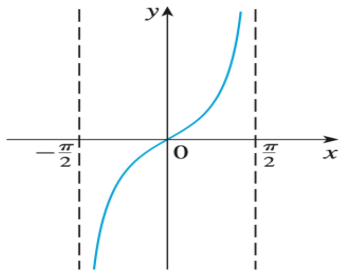
Definition 23 (The Inverse Function of $y = \sin x$): We consider $y = \sin x : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1]$. This is one-to-one function. Therefore, it has an inverse function, that is denoted by $y = \arcsin x : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$



Definition 24 (The Inverse Function of $y = \cos x$): We consider $y = \cos x : [0, \pi] \longrightarrow [-1, 1]$. This is one-to-one function. Therefore, it has an inverse function, that is denoted by $y = \arccos x : [-1, 1] \longrightarrow [0, \pi]$



Definition 25 (The Inverse Function of $y = \tan x$): We consider $y = \tan x : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow (-\infty, +\infty)$. This is one-to-one function. Therefore, it has an inverse function, that is denoted by $y = \arctan x : (-\infty, +\infty) \rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$



Example 28: Show that

$$\forall x \in [-1, 1], \quad \arcsin x + \arccos x = \frac{\pi}{2}$$

Solution : Let

$$a = \arcsin x, \quad b = \arccos x \Rightarrow x = \sin a = \cos b = \sin \left(\frac{\pi}{2} - b \right)$$

We obtain

$$a = \frac{\pi}{2} - b \Rightarrow a + b = \frac{\pi}{2}$$

Therefore

$$\arcsin x + \arccos x = \frac{\pi}{2}$$

Example 29: Show that

$$\forall x \in \mathbb{R}, \quad \arctan x + \arctan y = \arctan \frac{x + y}{1 - xy}$$

Solution : Let

$$a = \arctan x \Rightarrow x = \tan a \quad \text{and} \quad b = \arctan y \Rightarrow y = \tan b$$

We obtain

$$\tan(a + b) = \frac{\tan a + \tan b}{1 - \tan a \tan b} = \frac{x + y}{1 - xy}$$

Therefore

$$a + b = \arctan x + \arctan y = \arctan \frac{x + y}{1 - xy}$$

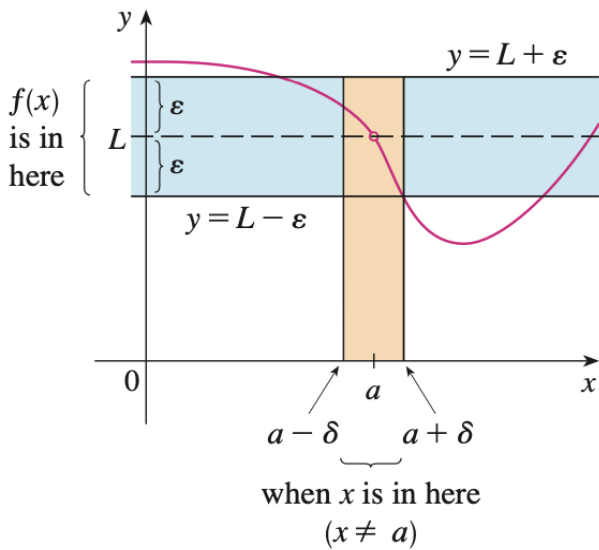
In a special case, if $y = \frac{1}{x}$ then $\arctan x + \arctan \frac{1}{x} = \pm \frac{\pi}{2}$.

We choose the (+) sign if $x > 0$ and the (−) sign if $x < 0$.

LIMITS OF A FUNCTION

Definition 26: Let f be a function defined on some open interval that contains the number a , except possibly at a itself. Then we say that the **limit** of $f(x)$ as x approaches a is L , and we write $\lim_{x \rightarrow a} f(x) = L$ if for every number $\varepsilon > 0$ there is a number $\delta > 0$ such that if $0 < |x - a| < \delta$ then $|f(x) - L| < \varepsilon$. That is

$$\forall \varepsilon > 0, \exists \delta > 0, 0 < |x - a| < \delta \Rightarrow |f(x) - L| < \varepsilon$$



Example 30: Show that $\lim_{x \rightarrow 2} (3x - 2) = 4$.

Solution : We have $a = 2$, $L = 4$, $f(x) = 3x - 2$ and we need to find a number $\delta > 0$ such that $\forall \varepsilon > 0$, if $0 < |x - 2| < \delta$ then $|(3x - 2) - 4| < \varepsilon$. From the last inequality, we have

$$\begin{aligned} |(3x - 2) - 4| < \varepsilon &\Leftrightarrow |3x - 6| < \varepsilon \Leftrightarrow |3(x - 2)| < \varepsilon \\ &\Leftrightarrow |x - 2| < \frac{\varepsilon}{3} \end{aligned}$$

This suggests that we should choose $\delta = \frac{\varepsilon}{3}$.

Theorem 2 (Algebraic Laws): Suppose that C is a constant and the limits $\lim_{x \rightarrow a} f(x)$ and $\lim_{x \rightarrow a} g(x)$ exist. Then

$$(1) \lim_{x \rightarrow a} [f(x) + g(x)] = \lim_{x \rightarrow a} f(x) + \lim_{x \rightarrow a} g(x)$$

$$(2) \lim_{x \rightarrow a} [f(x) - g(x)] = \lim_{x \rightarrow a} f(x) - \lim_{x \rightarrow a} g(x)$$

$$(3) \lim_{x \rightarrow a} [Cf(x)] = C \lim_{x \rightarrow a} f(x)$$

$$(4) \lim_{x \rightarrow a} [f(x)g(x)] = \lim_{x \rightarrow a} f(x) \cdot \lim_{x \rightarrow a} g(x)$$

$$(5) \lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} \quad \text{if } \lim_{x \rightarrow a} g(x) \neq 0$$

Theorem 3 (Limit of Composite Functions): *If*

$$\lim_{x \rightarrow b} f(x) = L \text{ and } \lim_{x \rightarrow a} g(x) = b, \text{ then } \lim_{x \rightarrow a} f(g(x)) = L$$

Example 31: Suppose that $\lim_{x \rightarrow 2} f(x) = 3$ and $\lim_{x \rightarrow 1} g(x) = 2$.
Then $\lim_{x \rightarrow 1} f(g(x)) = 3$.

NOTE:

From the above laws, we have, if $\lim_{x \rightarrow a} f(x) = L$, then

1. $\lim_{x \rightarrow a} [f(x)]^n = L^n$
2. $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{L} \quad (L > 0)$

Theorem 4 (Direct Substitution Property): *If f is a combination or composition of elementary functions and a is in the domain of f , then*

$$\lim_{x \rightarrow a} f(x) = f(a)$$

Example 32: Find $\lim_{x \rightarrow 1} f(x)$ where $f(x) = \frac{\sqrt[3]{x^2 + 2x - 7}}{\sqrt{3x^3 + 5x + 8}}$.
Since the function $f(x)$ is an algebraic function and $a = 1$ is in its domain, then

$$\lim_{x \rightarrow 1} f(x) = f(1) = \frac{\sqrt[3]{1^2 + 2 \cdot 1 - 7}}{\sqrt{3 \cdot 1^3 + 5 \cdot 1 + 8}} = \frac{\sqrt[3]{-4}}{\sqrt{16}} = -\frac{\sqrt[3]{4}}{4}$$

Theorem 5 (The Inequality Theorem): *If $f(x) \leq g(x)$ when x is near a and the limits of f and g both exist as x approaches a , then*

$$\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x)$$

Theorem 6 (The Squeeze Theorem):
Suppose

$$f(x) \leq g(x) \leq h(x)$$

when x is near a and if

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$$

then

$$\lim_{x \rightarrow a} g(x) = L$$

Example 33: Find $\lim_{x \rightarrow 0} x^2 \sin^2 \frac{1}{x}$.

Solution : We know that

$$0 \leq x^2 \sin^2 \frac{1}{x} \leq x^2$$

and

$$\lim_{x \rightarrow 0} x^2 = 0$$

By the Squeeze Theorem, we obtain

$$\lim_{x \rightarrow 0} x^2 \sin^2 \frac{1}{x} = 0$$

Definition 27 (Infinite Limits):

$$\lim_{x \rightarrow a} f(x) = +\infty \Leftrightarrow \forall M > 0, \exists \delta > 0, \text{ such that} \\ 0 < |x - a| < \delta \Rightarrow f(x) > M$$

$$\lim_{x \rightarrow a} f(x) = -\infty \Leftrightarrow \forall N < 0, \exists \delta > 0, \text{ such that} \\ 0 < |x - a| < \delta \Rightarrow f(x) < N$$

Definition 28 (Limits at Infinity): Let $f(x)$ be a function defined on some interval (a, ∞) . Then

$$\lim_{x \rightarrow \infty} f(x) = L \Leftrightarrow \forall \varepsilon > 0, \exists N > 0, \forall x > N \Rightarrow |f(x) - L| < \varepsilon$$

and

$$\lim_{x \rightarrow \infty} f(x) = \infty \Leftrightarrow \forall M > 0, \exists N > 0, \forall x > N \Rightarrow f(x) > M$$

Some useful limits

We consider some useful limits of the elementary functions when x approaches $\pm\infty$.

- $\alpha > 0 \implies \lim_{x \rightarrow +\infty} x^\alpha = +\infty$ and
 $\alpha < 0 \implies \lim_{x \rightarrow +\infty} x^\alpha = 0$
- If $a > 1$, then $\lim_{x \rightarrow +\infty} a^x = +\infty$ and $\lim_{x \rightarrow -\infty} a^x = 0$
- If $a > 1$, then $\lim_{x \rightarrow +\infty} \log_a x = +\infty$
- $$\lim_{x \rightarrow +\infty} \frac{a_0 x^n + a_1 x^{n-1} + \dots + a_n}{b_0 x^m + b_1 x^{m-1} + \dots + b_m} = \begin{cases} 0, & \text{if } n < m \\ \frac{a_0}{b_0}, & \text{if } n = m \\ \infty, & \text{if } n > m \end{cases}$$

- The trigonometric functions ($\sin x, \cos x, \tan x, \cot x$) have no limits when x approaches $\pm\infty$.
- We have

$$\lim_{x \rightarrow +\infty} \arctan x = \frac{\pi}{2} \quad \text{and} \quad \lim_{x \rightarrow -\infty} \arctan x = -\frac{\pi}{2}$$

Sometime, we can write

$$\arctan(+\infty) = \frac{\pi}{2} \quad \text{and} \quad \arctan(-\infty) = -\frac{\pi}{2}$$

- $\lim_{x \rightarrow \pm\infty} \cosh x = +\infty, \quad \lim_{x \rightarrow \pm\infty} \sinh x = \pm\infty,$
 $\lim_{x \rightarrow +\infty} \tanh x = 1, \quad \lim_{x \rightarrow -\infty} \tanh x = -1$

ONE-SIDE LIMITS

Definition 29 (The Left-Hand Limit):

$$\lim_{x \rightarrow a^-} f(x) = L \Leftrightarrow \forall \varepsilon > 0, \exists \delta > 0, \text{ such that}$$
$$a - \delta < x < a \Rightarrow |f(x) - L| < \varepsilon$$

Definition 30 (The Right-Hand Limit):

$$\lim_{x \rightarrow a^+} f(x) = L \Leftrightarrow \forall \varepsilon > 0, \exists \delta > 0, \text{ such that}$$
$$a < x < a + \delta \Rightarrow |f(x) - L| < \varepsilon$$

Example 34: Consider the function $f(x) = \frac{1}{x}$. We have

$$\lim_{x \rightarrow 0^+} \frac{1}{x} = +\infty \quad \text{and} \quad \lim_{x \rightarrow 0^-} \frac{1}{x} = -\infty$$

Example 35: Consider the function

$$f(x) = \begin{cases} 1 - x, & \text{if } x < 1 \\ x^2, & \text{if } x \geq 1 \end{cases}$$

We have

$$\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} (1 - x) = 0 \quad \text{and} \quad \lim_{x \rightarrow 1^+} f(x) = \lim_{x \rightarrow 1^+} x^2 = 1$$

Example 36: Consider the function $f(x) = \frac{1}{2^{\frac{1}{x}} + 1}$.

- When x approaches 0 from the left, we have $\frac{1}{x} \longrightarrow -\infty$, then $2^{\frac{1}{x}} \longrightarrow 0$, so that $2^{\frac{1}{x}} + 1 \longrightarrow 1$. This means that $\lim_{x \rightarrow 0^-} f(x) = 1$.
- On the other hand, when x approaches 0 from the right, we have $\frac{1}{x} \longrightarrow +\infty$, then $2^{\frac{1}{x}} \longrightarrow +\infty$, and $2^{\frac{1}{x}} + 1 \longrightarrow +\infty$. This means that $\lim_{x \rightarrow 0^+} f(x) = 0$.

Theorem 7:

$$\lim_{x \rightarrow a} f(x) = L \text{ if and only if } \lim_{x \rightarrow a^-} f(x) = L = \lim_{x \rightarrow a^+} f(x).$$

NOTE:

From the Direct Substitution Property we see that if a function $f(x)$ is defined at a number a then the limit of this function when x approaches a can be found by replacing x by a . Therefore, in reality, we often encounter with limits of functions which are not defined at a . In these cases, we need to apply other tools to calculate these limits.

Example 37: Find $\lim_{x \rightarrow 2} \frac{x^2 - 3x + 2}{x - 2}$.

Solution : The function $f(x) = \frac{x^2 - 3x + 2}{x - 2}$ is not defined at $x = 2$, so we can not replace x by 2. In order to find the limit we factor the numerator as $x^2 - 3x + 2 = (x - 1)(x - 2)$ and

$$\lim_{x \rightarrow 2} \frac{x^2 - 3x + 2}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 1)(x - 2)}{x - 2} = \lim_{x \rightarrow 2} (x - 1) = 1$$

In the second expression we can simplify the term $x - 2$ because although x approaches to 2 but x differs from 2. That means $x - 2 \neq 0$.

Definition 31: We used to find the limits of following **indefinite forms**:

$$\frac{0}{0}, \quad \frac{\infty}{\infty}, \quad \infty - \infty, \quad 1^{\infty}, \quad 0^{\infty}, \quad 0 \cdot \infty, \quad \dots$$

SOME USEFUL LIMITS

$$(1) \lim_{x \rightarrow 0} \frac{\sin x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$$

$$(2) \lim_{x \rightarrow 0} \frac{e^x - 1}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\ln(1 + x)}{x} = 1$$

$$(3) \lim_{x \rightarrow 0} \frac{(1 + x)^{\alpha} - 1}{x} = \alpha$$

$$(4) \lim_{x \rightarrow 0} \frac{\sinh x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{\tanh x}{x} = 1, \quad \lim_{x \rightarrow 0} \frac{1 - \cosh x}{x^2} = -\frac{1}{2}$$

SOME USEFUL LIMITS

$$(5) \quad \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e, \quad \lim_{x \rightarrow 0} (1 + x)^{\frac{1}{x}} = e$$

$$(6) \quad \lim_{x \rightarrow \infty} \frac{a_0 x^n + a_1 x^{n-1} + \cdots + a_n}{b_0 x^m + b_1 x^{m-1} + \cdots + b_m} = \begin{cases} 0, & \text{if } n < m \\ \frac{a_0}{b_0}, & \text{if } n = m \\ \infty, & \text{if } n > m \end{cases}$$

Example 38: Find $\lim_{x \rightarrow 0} \frac{\sin 7x}{\tan 3x}$. We can rewrite

$$\lim_{x \rightarrow 0} \frac{\sin 7x}{\tan 3x} = \lim_{x \rightarrow 0} \frac{\sin 7x}{7x} \cdot \frac{3x}{\tan 3x} \cdot \frac{7}{3} = \frac{7}{3}$$

Example 39: Find $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$. We can rewrite

$$\begin{aligned}\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} &= \lim_{x \rightarrow 0} \frac{\frac{\sin x}{\cos x} - \sin x}{x^3} \\&= \lim_{x \rightarrow 0} \frac{\sin x - \sin x \cos x}{x^3 \cos x} \\&= \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{1 - \cos x}{x^2} \cdot \frac{1}{\cos x} \\&= 1 \cdot \frac{1}{2} \cdot \frac{1}{1} = \frac{1}{2}\end{aligned}$$

Example 40: Find $\lim_{x \rightarrow +\infty} x^{3/2} (\sqrt{x^3 + 2} - \sqrt{x^3 - 2})$. We have

$$\begin{aligned}\lim_{x \rightarrow +\infty} f(x) &= \lim_{x \rightarrow +\infty} x^{3/2} \frac{(x^3 + 2) - (x^3 - 2)}{\sqrt{x^3 + 2} + \sqrt{x^3 - 2}} \\&= \lim_{x \rightarrow +\infty} \frac{4\sqrt{x^3}}{\sqrt{x^3 + 2} + \sqrt{x^3 - 2}} \\&= \lim_{x \rightarrow +\infty} \frac{4}{\sqrt{1 + \frac{2}{x^3}} + \sqrt{1 - \frac{2}{x^3}}} = \frac{4}{2} = 2\end{aligned}$$

Example 41: Find $\lim_{x \rightarrow \infty} \left(\frac{x+2}{x-2} \right)^x$.

$$\begin{aligned} \lim_{x \rightarrow \infty} \left(\frac{x+2}{x-2} \right)^x &= \lim_{x \rightarrow \infty} \left(\frac{x-2+4}{x-2} \right)^x \\ &= \lim_{x \rightarrow \infty} \left(1 + \frac{4}{x-2} \right)^x = \lim_{x \rightarrow \infty} \left(1 + \frac{1}{\frac{x-2}{4}} \right)^x \\ &= \lim_{x \rightarrow \infty} \left(1 + \frac{1}{\frac{x-2}{4}} \right)^{\frac{x-2}{4} \cdot \frac{4}{x-2} \cdot x} = \lim_{x \rightarrow \infty} \left[\left(1 + \frac{1}{\frac{x-2}{4}} \right)^{\frac{x-2}{4}} \right]^{\frac{4x}{x-2}} \\ &= e^4 \end{aligned}$$

Example 42: Find $\lim_{x \rightarrow 0} (\cos x)^{\frac{1}{x^2}}$. We have

$$\begin{aligned}\lim_{x \rightarrow 0} (\cos x)^{\frac{1}{x^2}} &= \lim_{x \rightarrow 0} (1 + \cos x - 1)^{\frac{1}{x^2}} \\ &= \lim_{x \rightarrow 0} \left[(1 + (\cos x - 1))^{\frac{1}{\cos x - 1}} \right]^{\frac{\cos x - 1}{x^2}} \\ &= e^{-1/2}\end{aligned}$$

ASYMPTOTES

Definition 32: The line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following statements is true

$$\lim_{x \rightarrow a} f(x) = \infty$$

$$\lim_{x \rightarrow a^-} f(x) = \infty$$

$$\lim_{x \rightarrow a^+} f(x) = \infty$$

$$\lim_{x \rightarrow a} f(x) = -\infty$$

$$\lim_{x \rightarrow a^-} f(x) = -\infty$$

$$\lim_{x \rightarrow a^+} f(x) = -\infty$$

Definition 33: The line $y = L$ is called a **horizontal asymptote** of the curve $y = f(x)$ if either

$$\lim_{x \rightarrow \infty} f(x) = L$$

or

$$\lim_{x \rightarrow -\infty} f(x) = L$$

Example 43: Find vertical and horizontal asymptotes of the function

$$f(x) = \frac{2x^2 + 3}{x^2 - 1}$$

Solution : We have

$$\lim_{x \rightarrow \infty} f(x) = \lim_{x \rightarrow \infty} \frac{2x^2 + 3}{x^2 - 1} = 2$$

Then, the horizontal asymptote is $y = 2$. And

$$\lim_{x \rightarrow \pm 1} f(x) = \lim_{x \rightarrow \pm 1} \frac{2x^2 + 3}{x^2 - 1} = \pm \infty$$

Therefore, the vertical asymptotes are $x = -1$ and $x = 1$.

Definition 34: The line $y = mx + b$ is called a **slant asymptote** of the curve $y = f(x)$ if there exist the following limits

$$m = \lim_{x \rightarrow \infty} \frac{f(x)}{x}, \quad b = \lim_{x \rightarrow \infty} [f(x) - mx]$$

Example 44: Find all asymptotes of the curve $y = \frac{2x^2 + 1}{x - 2}$.

Solution : First of all, we have

$$\lim_{x \rightarrow 2} f(x) = \lim_{x \rightarrow 2} \frac{2x^2 + 1}{x - 2} = \pm \infty$$

Therefore, the vertical asymptote is $x = 2$.

We see that

$$m = \lim_{x \rightarrow \infty} \frac{f(x)}{x} = \frac{2x^2 + 1}{x(x - 2)} = 2$$

$$\begin{aligned} b &= \lim_{x \rightarrow \infty} [f(x) - mx] = \lim_{x \rightarrow \infty} \left[\frac{2x^2 + 1}{x - 2} - 2x \right] \\ &= \lim_{x \rightarrow \infty} \frac{(2x^2 + 1) - 2x(x - 2)}{x - 2} = \lim_{x \rightarrow \infty} \frac{4x + 1}{x - 2} = 4 \end{aligned}$$

So, the slant asymptote is $y = 2x + 4$.

INFINITESIMALS

Definition 35: A function $\alpha(x)$ is called an **infinitesimal** when x approaches a if

$$\lim_{x \rightarrow a} \alpha(x) = 0$$

Example 45:

- The functions x^2 , $\sin x$, $\tan x$, $1 - \cos x$, ... are infinitesimals when $x \rightarrow 0$.
- The functions $\frac{1}{x}$, 2^{-x} , ... are infinitesimals when $x \rightarrow +\infty$.
- The function $x^3 - 1$, $\cot \frac{\pi x}{2}$, ... are infinitesimals when $x \rightarrow 1$.

Definition 36: Two infinitesimals $\alpha(x)$ and $\beta(x)$ when x approaches a are called **equivalent** if

$$\lim_{x \rightarrow a} \frac{\alpha(x)}{\beta(x)} = 1$$

In this case, we denote

$$\alpha(x) \sim \beta(x) \quad (x \rightarrow a)$$

Example 46:

- $\sin x \sim x$ ($x \rightarrow 0$), since $\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$
- $1 - \cos x \sim \frac{x^2}{2}$ ($x \rightarrow 0$), since $\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}$

We have some useful equivalent infinitesimals when x approaches 0

$$\left\{ \begin{array}{l} \sin x \sim x - \frac{x^3}{6} \\ \tan x \sim x + \frac{x^3}{3} \\ 1 - \cos x \sim \frac{x^2}{2} - \frac{x^4}{24} \\ e^x - 1 \sim x + \frac{x^2}{2} + \frac{x^3}{6} \\ (1+x)^\alpha - 1 \sim \alpha x + \frac{\alpha(\alpha-1)}{2}x^2 \end{array} \right.$$

Definition 37: Given two infinitesimals $\alpha(x)$ and $\beta(x)$ when x approaches a . We say that $\alpha(x)$ is an **infinitesimal of higher degree** in comparison to $\beta(x)$, if

$$\lim_{x \rightarrow a} \frac{\alpha(x)}{\beta(x)} = 0$$

In this case we denote

$$\alpha(x) = o(\beta(x)), \quad (x \rightarrow a)$$

Theorem 8 (Replacement by Equivalent Infinitesimals):

Given two infinitesimals $f(x)$ and $g(x)$ when x approaches a . Suppose that $f(x) \sim \alpha(x)$, $g(x) \sim \beta(x)$, ($x \rightarrow a$), and

$$\lim_{x \rightarrow a} \frac{\alpha(x)}{\beta(x)} = L. \text{ Then}$$

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = L$$

Example 47: Find $\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3}$.

We know that

$$\tan x \sim x + \frac{x^3}{3}, \quad \sin x \sim x - \frac{x^3}{6}, \quad (x \rightarrow 0)$$

Therefore,

$$\tan x - \sin x \sim \left(x + \frac{x^3}{3}\right) - \left(x - \frac{x^3}{6}\right) = \frac{x^3}{2}, \quad (x \rightarrow 0)$$

Then

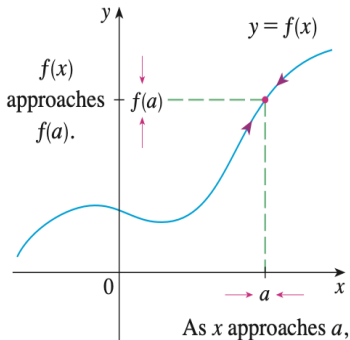
$$\lim_{x \rightarrow 0} \frac{\tan x - \sin x}{x^3} = \lim_{x \rightarrow 0} \frac{x^3/2}{x^3} = \frac{1}{2}$$

CONTINUITY

Definition 38: A function f is **continuous** at a if $\lim_{x \rightarrow a} f(x) = f(a)$. Otherwise, we say that f is **discontinuous** at a .

This definition requires three things if f is continuous at a :

- (1) f is defined at a
- (2) $\lim_{x \rightarrow a} f(x)$ exists
- (3) $\lim_{x \rightarrow a} f(x) = f(a)$



Example 48: Given $f(x) = \begin{cases} \frac{x^2 - 4}{x - 2}, & \text{if } x \neq 2 \\ A, & \text{if } x = 2 \end{cases}$. Find A such that f is continuous at $x = 2$.

Solution : We see that f is defined at $x = 2$, $f(2) = A$ and

$$\begin{aligned} \lim_{x \rightarrow 2} f(x) &= \lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2} = \lim_{x \rightarrow 2} \frac{(x - 2)(x + 2)}{x - 2} \\ &= \lim_{x \rightarrow 2} (x + 2) = 4 \end{aligned}$$

Note that in the last expression we can simplify the term $x - 2$ since when x approaches 2 means that $x \neq 2$. And from the above definition, f is continuous at $x = 2$ if $A = 4$.

Definition 39: We have the following definitions

- (1) A function f is **continuous from the right** at a if

$$\lim_{x \rightarrow a^+} f(x) = f(a)$$

- (2) A function f is **continuous from the left** at a if

$$\lim_{x \rightarrow a^-} f(x) = f(a)$$

- (3) A function f is **continuous on an opened interval** (a, b) if it is continuous at every number in the interval.

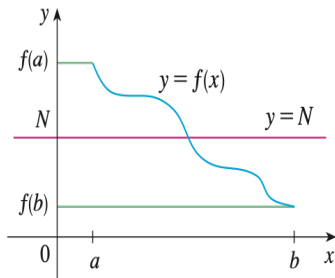
- (4) A function f is **continuous in a close interval** $[a, b]$ if it is continuous on (a, b) , continuous from the right at a , and continuous from the left at b .

Properties of Continuous Functions

- (1) A function f is continuous at a when and only when it is continuous from the left and from the right at a .
- (2) The addition, subtraction, multiplication, division, and composition of continuous functions are continuous functions.
- (3) The elementary functions are continuous on their domains.
- (4) The continuous function $f(x)$ in a close interval $[a, b]$ attains its maximum and minimum values in $[a, b]$.
This means that there exist two numbers c and d in $[a, b]$ such that $f(c) \leq f(x) \leq f(d), \forall x \in [a, b]$.

Theorem 9 (The Intermediate Value Theorem): Suppose f is continuous on the closed interval $[a, b]$ and let N be any number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then there exists a number c in (a, b) such that $f(c) = N$.

This theorem states that a continuous function takes on every intermediate value between the function values $f(a)$ and $f(b)$. It is illustrated in the figure on the right. Note that the value can be taken on once or more than once.



Example 49: Show that there is a root of the equation $f(x) = x^3 - 3x + 1 = 0$ between 1 and 2.

Solution : We are looking for a solution of the given equation, that is, a number c between 1 and 2 such that $f(c) = 0$. Therefore we take $a = 1$, $b = 2$, and $N = 0$ in this theorem. We have $f(1) < 0 < f(2)$; that is, $N = 0$ is a number between $f(1)$ and $f(2)$. Now f is continuous since it is a polynomial, so the Intermediate Value Theorem says there is a number c between 1 and 2 such that $f(c) = 0$. In other words, the equation $x^3 - 3x + 1 = 0$ has at least one root in the interval $(1, 2)$.

— THE END —